

EMAuxiliary Reference Manual

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Getting started

Make sure you have completed the following setup steps:

1. **Install Blimp** (a free, standalone program):
Download from <https://www.appliedmissingdata.com/blimp>
2. **Install the R interface for Blimp (rblimp)**: run this in your R environment:
`remotes::install_github("blimp-stats/rblimp")`
3. **Import the EMAuxiliary** function by sourcing it directly from GitHub. Run this in R:
`source("https://raw.githubusercontent.com/schneids111/EMAuxiliary-Workflow/main/functions/EMAuxiliary.R")`
4. **Prepare the EMA dataset** by adding rows for missing prompts and appropriately aligning auxiliary variables with observed and missing prompts in the data

EMAuxiliary function

The code below shows the required and optional elements of the *EMAuxiliary* function.

```
results <- EMAuxiliary(  
  y ~ x + m + x:m + (1 + x | id),      # focal model using lme4-style syntax  
  data = dat,                          # dataset  
  id = "id",                           # subject ID / clustering variable  
  aux = c("aux1", "aux2", "aux3"),     # list of auxiliary variables to include  
  center_group = x,                    # optional: within-person centering  
  center_grand = m,                    # optional: grand-mean centering  
  ordinal = "aux2",                     # optional: ordinal variable  
  nominal = "aux3",                     # optional: nominal (unordered categorical) variable  
  burn = 10000,                         # optional: number of burn-in iterations  
  iter = 10000,                         # optional: number of post-burn-in iterations  
)
```

EMAuxiliary Line-by-Line Walkthrough

Focal Model ($y \sim x + m + x:m + (1 + x | id)$):

This line specifies your multilevel model using lme4 syntax. Fixed effects and interactions can be written as $x_1 + x_2 + x_1:x_2$ or more compactly as $x_1 * x_2$. Random effects follow the usual lme4 format, such as $(1 + x | id)$ for a model with a random intercept and random slope of x .

Dataset (`data = dat`):

Specifies the dataset.

Clustering Variable (`id = "id"`):

Specifies the subject identifier for clustering. EMAuxiliary supports two-level models (e.g.,

observations within persons) with any number of within- or between-person effects and interactions, including cross-level interactions. For three-level models, users must work directly in Blimp.

Auxiliary Variables (`aux = c(...)`):

Auxiliary variables are specified as additional multivariate dependent variables (DVs) in the multilevel models. EMAuxiliary automatically determines the level of variation for each auxiliary variable, and regresses them on the corresponding components of the focal model using the following rules:

- **Within-person auxiliaries** (e.g., time of day) are regressed on the within-person components of the outcome and predictor variables.
- **Between-person auxiliaries** (e.g., age) are regressed on the latent between-person components of the outcome and predictor variables.
- **Mixed auxiliaries with variation on both levels** are regressed on both within- and between-person components, including possible cross-level interactions.

This is handled automatically—no manual processing of auxiliary variables is required.

Centering Predictors (`center_group`, `center_grand`):

Users can optionally specify group-mean and grand-mean centering for predictors:

- `center_group = x` centers `x` within person.
- `center_grand = x` centers the person-level mean of `x` around the grand mean.

Vectors of predictors can also be centered [e.g., `center_group = c("x1", "x2")`], not just individual predictors.

Auxiliary variables do *not* need to be centered, as they enter the model as multilevel outcome (not predictor) variables.

Special Feature: Latent Mean Centering

The `center_group` option in *EMAuxiliary* activates latent within-person centering in Blimp. This feature improves the estimation of within- and between-person effects, especially in the presence of missing data.

When a predictor is listed under `center_group`, Blimp automatically decomposes it into:

- a latent within-person component, representing deviations from the individual's latent mean, and
- a latent between-person component, representing the individual's latent mean across all occasions.

Both components can then be included as predictors in the multilevel model. The between-person component is referenced as `x.mean` in the model equation. For example:

- `center_group = x`
combined with
- `y ~ x + x.mean + (1 + x | id)`

regresses the outcome `y` on the latent within-person part of `x` (`x`) and the latent between-person part of `x` (`x.mean`), estimating a random intercept for `y` and a random slope for the effect of `x` on `y`.

Variable Types (ordinal, nominal):

Non-continuous variables can be declared using the following arguments:

- ordinal = "aux2" — for binary or ordered categorical variables (e.g., gender, education).
- nominal = "aux3" — for unordered categorical variables (e.g., race, region).

Any variable—whether predictor, outcome, or auxiliary—can be specified as ordinal or nominal. Vectors of ordinal and nominal variables can also serve as input, not just individual variables.

When an ordinal variable is specified, Blimp estimates probit regression models appropriate for binary or ordered outcomes.

When a nominal variable is specified, Blimp estimates multinomial logistic regression models. Nominal variables are automatically dummy-coded, using the lowest category as the reference group. Note that nominal variables with many categories may substantially increase computational burden.

Bayesian Estimation Settings (burn, iter) and Model Convergence:

Blimp uses Bayesian estimation based on Markov Chain Monte Carlo (MCMC) sampling. Two key parameters control the sampling process:

- burn – Number of initial samples to discard (the *burn-in* period).
- iter – Number of samples to retain after burn-in.

Model convergence is evaluated using the Potential Scale Reduction (PSR) statistic. A PSR value greater than 1.05 suggests that the chains have not mixed well and that additional iterations may be needed. Increasing the burn-in period (e.g., doubling the default value of 10,000) often helps the model reach convergence. Models with larger numbers of auxiliary variables and more iterations will take longer processing times. It is worth the wait!

MCMC algorithms generally sample more efficiently when variables are on comparable scales—typically with variances around 1.0. Large differences in variable scaling can slow mixing and hinder convergence. *EMAuxiliary* automatically standardizes auxiliary variables to stabilize estimation, but large variances in the main model variables (predictors or outcomes) can still cause convergence problems. When detected, *EMAuxiliary* issues a warning and suggests rescaling those variables.

Viewing and Interpreting Output

After model estimation, EMAuxiliary provides several options for inspecting results at different levels of detail:

```
cat(results$blimp_code)    # show generated BLIMP program code
blimp_print_psr(results)   # PSR section only
blimp_print_focal(results) # outcome section for focal analysis model only
rblimp::output(results$fit) # full output
```

cat(results\$blimp_code)

Displays the Blimp syntax created by EMAuxiliary.

Useful for:

- verifying variable names and centering options
- confirming that latent mean centering is correctly applied
- debugging or re-running the model directly in Blimp

blimp_print_psr(results)

Prints the Potential Scale Reduction (PSR) summary section only, allowing quick checks of model convergence. Blimp prints a *PSR value at multiple points during sampling*, showing how convergence improves as the chains mix. The sequence of PSR values reflects the chain's progress, and only the final PSR value at the bottom of the PSR section should be used to judge convergence.

Interpreting the final PSR value:

- **PSR < 1.05 — Good convergence**
Chains have mixed well; no further action needed.
- **1.05 ≤ PSR < 1.10 — Borderline**
Convergence is close but not optimal. Increasing *burn-in* usually resolves this.
- **PSR ≥ 1.10 — Convergence failure**
The chains did not mix sufficiently. Scaling issues, problematic variables, or model misspecification may be contributing.
- **PSR ≥ 1.20 — Serious convergence problems**
Indicates potential substantial instability. Model structure or variable scaling should be closely reviewed before simply adding more iterations.

Example output:

Comparing iterations across 2 chains	Highest PSR	Parameter #
126 to 250	1.311	11
251 to 500	1.283	11
376 to 750	1.158	11
501 to 1000	1.085	10
626 to 1250	1.160	11
751 to 1500	1.094	11
876 to 1750	1.230	11
1001 to 2000	1.266	11
1126 to 2250	1.267	11
1251 to 2500	1.186	11
1376 to 2750	1.136	11
1501 to 3000	1.096	11
1626 to 3250	1.084	11
1751 to 3500	1.049	11
1876 to 3750	1.039	11
2001 to 4000	1.022	5
2126 to 4250	1.027	11
2251 to 4500	1.009	7
2376 to 4750	1.007	18
2501 to 5000	1.024	11

Reminder: Blimp prints a *series* of PSR values, but the only value that matters is the one at the bottom of the list, which summarizes convergence after all burn-in and sampling iterations.

blimp_print_focal(results)

Shows the focal model output—that is, the parameter estimates for the primary outcome(s) and predictors, excluding auxiliary-variable regressions.

Example output is shown below:

Parameters	Estimate	StdDev	2.5%	97.5%	ChiSq	PValue	N_Eff

Variances:							
L2 : Var(Intercept)	15.090	2.780	10.568	21.259	---	---	964.706
L2 : Cov(x, Intercept)	-0.770	0.828	-2.407	0.802	---	---	146.303
L2 : Var(x)	0.532	0.470	0.088	1.813	---	---	41.832
Residual Var.	10.291	0.908	8.687	12.272	---	---	455.118
Coefficients:							
Intercept	25.080	0.438	24.213	25.908	3283.133	0.000	175.502
x	0.082	0.191	-0.290	0.453	0.176	0.675	558.309
x.mean[id]	-0.053	0.620	-1.449	1.026	0.021	0.884	67.538
x*x.mean[id]	-0.767	0.266	-1.328	-0.275	8.355	0.004	339.709
Standard Deviations:							
L2 : SD(Intercept)	3.885	0.351	3.251	4.611	---	---	952.475
L2 : Cor(x, Intercept)	-0.271	0.290	-0.861	0.254	---	---	47.293
L2 : SD(x)	0.730	0.276	0.297	1.346	---	---	32.501
Residual SD	3.208	0.141	2.947	3.503	---	---	450.324
Standardized Coefficients:							
x	0.018	0.043	-0.066	0.102	0.175	0.675	561.464
x.mean[id]	-0.009	0.098	-0.223	0.163	0.019	0.889	68.783
x*x.mean[id]	-0.143	0.047	-0.234	-0.051	9.205	0.002	440.483
Proportion Variance Explained							
by Coefficients	0.031	0.019	0.007	0.077	---	---	245.278
by Level-2 Random Intercepts	0.555	0.050	0.451	0.649	---	---	762.423
by Level-2 Random Slopes	0.027	0.023	0.005	0.090	---	---	39.669
by Level-1 Residual Variation	0.377	0.047	0.293	0.473	---	---	468.789

The focal model output mirrors what researchers are used to from mixed-effects modeling (e.g., lme4), but with additional information from Bayesian estimation. Below is a brief guide to the key columns and sections.

a) Columns

- Estimate: The estimated effect.
For fixed effects: interpret like a regression coefficient (e.g., effect of x on y).
For variances: the estimated variability of the random intercepts, random slopes, or residuals.
- StdDev: Uncertainty in the estimate (analogous to a standard error).
- 2.5% and 97.5%: A 95% interval for the parameter (a “credible interval”, analogous to a confidence interval).
If this interval excludes zero → evidence the effect is credibly non-zero (i.e., significant).
- ChiSq and PValue: Approximate significance tests included for convenience.
- N_Eff: Effective sample size. Low N_Eff (e.g., <50) suggests the parameter was harder to estimate. Can be improved by increasing the value of *iter*.

b) Sections

- Variances: Random-effect variances and covariances at Level 2, plus the Level-1 residual variance.
These correspond directly to random-effects output in frequentist (e.g., *lme4*) models.
- Coefficients: Fixed effects for:
 - the within-person effect of x (listed simply as x)
 - the between-person effect ($x.mean[id]$, created automatically when using `center_group`)
 - cross-level interactions (e.g., $x*x.mean[id]$)
 - the intercept.
 Interpretation is the same as in frequentist mixed models.
- Standard Deviations / Correlations: Square roots of the Level-2 variances and correlations among random effects.
Displayed for convenience; structurally the same as *lme4*’s “*Std.Dev*” and “*Corr*” portions of the random-effects table.
- Standardized Coefficients: Standardized versions of the fixed effects. Useful when predictors differ in scale.
- Proportion Variance Explained: Multilevel variance decomposition (similar to R^2 components), showing the proportion of total variability attributable to fixed effects, Level-2 random intercepts, Level-2 slopes, and Level-1 residuals.

rblimp::output(results\$fit)

Produces the complete multivariate output, equivalent to a Blimp .out file.

Includes:

- focal model results
- auxiliary-variable regressions
- covariance structures
- prior specifications
- PSR and chain diagnostics
- posterior summaries for all parameters

This is especially useful for advanced users who want to examine the full multivariate model or troubleshoot convergence.

Troubleshooting Tips

1. Data Formatting Issues

1.1 Categorical variables must be numeric

Blimp requires categorical variables to be represented with numeric codes. Before running *EMAuxiliary()*, ensure that variables such as gender, race, or region are stored as numeric values (e.g., 1, 2, 3). Character strings or factors should be converted prior to analysis. Use the ordinal or nominal arguments to indicate the appropriate variable type.

1.2 Pure between-person variables must be constant and complete within person

Variables that represent between-person characteristics (e.g., age, gender, baseline scores) must have the same value on every row for a given person. It is acceptable for a person to have all missing values for such a variable, but intermittent missing values within a person must be corrected by filling forward/backward to ensure constancy.

2. Auxiliary Variable Guidance

2.1 Auxiliary variables must not duplicate predictors or outcomes

Auxiliary variables should provide additional information about missingness. They should not duplicate variables already used in the focal model. Using the same variable (or a simple transformation of it) as both a predictor/outcome and an auxiliary adds no information and can create non-identifiable regression paths, leading to convergence problems.

Do not include as auxiliaries:

a) Variables already used as predictors

If a variable appears in the focal model (e.g., x in $y \sim x + m + (1 + x | id)$), it should not be listed as an auxiliary.

Common examples:

- `time_of_day` used as both a predictor and auxiliary
- `day` (study day/calendar day) used as both a predictor and auxiliary

b) The outcome variable

If y is the dependent variable, do not include: `aux = "y"`

c) Derived person-level means of predictors or outcomes

Person means such as: `x_mean`, `y_mean`, or any manually computed participant-level average of a focal variable should not be included as auxiliaries.

These are fully redundant with the model's latent between-person components and can readily contribute to identification or convergence issues.

A common misconception is that person means are as useful as summary auxiliary metrics, similar to derived variables of intraindividual variability (e.g., ISD, RMSSD).

They are not: variability measures contribute unique within-person information; person means do not.

3. Convergence Issues

3.1 Start with the default MCMC settings

We recommend beginning with the default settings (burn = 10000, iter = 10000). Using very small values for these settings to expedite runtime can hinder convergence and lead to misleading results.

If the final PSR value exceeds 1.05, the first step is to increase the burn-in period (e.g., 15000 or 20000). This typically allows the Markov chains more time to stabilize.

3.2 Interpreting PSR

The Potential Scale Reduction (PSR) statistic provides a quick assessment of convergence:

- **PSR < 1.05:** good convergence
- **1.05–1.20:** not considered good convergence but moderately increasing the value of “burn” is likely to help.
- **> 1.20:** suggests that burn and iter values may need to be substantially increased or the model needs to be reconsidered: could indicate a problematic variable, scaling issue, or misspecified model component

3.3 Rescaling predictors or outcomes

Blimp mixes most efficiently when variables have variances roughly near 1. Variables listed under *aux* are automatically standardized by *EMAuxiliary* before entering the models. For outcome and predictor variables in the focal equation, a warning is issued if $PSR > 1.05$ and if *EMAuxiliary* detects unusually large or small variances. In this case, consider z-transforming or rescaling focal variables by dividing or multiplying by a constant (e.g., dividing by 10, 100, or 1000).

3.4 Nominal variables with many categories

Nominal variables with numerous categories can lead to slow estimation and may contribute to convergence problems. As a practical guideline, nominal variables with more than five to 10 categories can be difficult to estimate in multivariate auxiliary models. Consider collapsing sparse categories or reconsidering whether the variable is essential.

4. Variable Referencing Errors

4.1 “Variable(s) do not exist in the dataset”

This error occurs when variables referenced in the model formula or other *EMAuxiliary* arguments are not present in the dataset.

Common causes include:

- typographical errors
- variables removed during preprocessing
- listing a variable in *aux*, *center_group*, *center_grand*, *ordinal*, or *nominal* that is not found in the data

Ensure that all variable names match exactly those in your dataset.

4.2 Using .mean incorrectly

Variables ending in *.mean* refer to latent between-person means, generated internally by Blimp when latent mean centering is enabled. These variables should never appear in the dataset and should not be created manually. If you reference *x.mean* in the formula, you must list “x” under *center_group*.

5. Data and Variable Characteristics

5.1 Variables that are near constants

If a variable is a near constant (e.g., the same value for all participants), it may not be used effectively as a predictor or auxiliary variable. Such variables typically lead to estimation difficulties or prevent convergence. Remove variables that have no observed variance before fitting the model.

5.2 Extremely sparse variables

Auxiliary variables with very high proportions of missingness (e.g., >90–95%) can lead to unstable estimation or poor convergence. Consider removing such variables unless they provide essential information.

6. Modeling Constraints

6.1 Random slopes cannot include latent mean variables

Random effects may only be specified for within-person variables.

Expressions such as: (1 + x.mean | id) are not permitted because x.mean is a between-person variable. Use only raw (within-person) variables in the random slope structure.

7. Understanding Messages, Warnings, and Errors from EMAuxiliary

EMAuxiliary produces diagnostic messages during preprocessing and model setup.

These fall into three categories:

- Informational messages — normal behavior; no action required
- Warnings — potential issues that may affect convergence
- Errors — the function stops and must be corrected before proceeding

Below are the key messages users will encounter.

7.1 “Standardized continuous auxiliaries: ...” (*informational*)

EMAuxiliary automatically standardizes continuous auxiliary variables to improve numerical stability. This is expected behavior and does not require user intervention.

7.2 “Potential cause for poor PSR: large scale differences among focal variables” (*warning*)

This warning appears when predictors or outcomes have very different scales (e.g., SDs of 1 vs. 1000), which can slow convergence.

Recommended action: Divide or multiply the affected variable(s) by a constant to place variances in a more reasonable range. This preserves substantive meaning while improving convergence.

7.3 “You used ‘x.mean’ but did not include ‘x’ in center_group” (*error; model will not run*)

This error is triggered when the model formula references a latent mean term (e.g., x.mean) without listing the corresponding base variable (x) in center_group. Because Blimp only creates latent mean variables for predictors specified under center_group, the function must stop.

Recommended action:

Either:

- add the base variable to center_group = c("x"), or

- remove x.mean from the formula.

7.4 “Auxiliary variable not found in the data” (*error; model will not run*)

This error indicates that a variable specified in aux, ordinal, nominal, center_group, or center_grand is not present in the dataset. *EMAuxiliary* cannot proceed because the requested variable is required for model construction.

Recommended action:

Correct the spelling/capitalization and ensure the variable is present in your dataset.

7.5 “The following variable(s) appear BOTH in the focal model formula and in the list of auxiliary variables” (*error; model will not run*)

This error indicates that one or more variables were entered as outcome or predictor in the focal regression equation and also in the list of auxiliary variables. *EMAuxiliary* will not proceed because this changes the model structure and produces unintended reciprocal regressions in the overall model.

Recommended action:

Remove the variable(s) listed in the error message from the 'aux' list.